2017 International Conference on Security, Pattern Analysis, and Cybernetics (SPAC)

Attribute Weighted Naive Bayes for Remote Sensing Image Classification Based on Cuckoo Search Algorithm

Juan Yang, Zhiwei Ye*, Xu Zhang, Wei Liu, Huazhong Jin
School of Computer Science
Hubei University of Technology
Wuhan, China
yangjuanline@163.com

***Abstract*—The Naive Bayes classifier(NB) is an effective and simple classification method for remote sensing image classification which is based on probability theory. However, in general, the contribution of each feature is different for classification and its attribute independence assumption is often invalid in the real world. The attribute weighted Naive Bayes(WNB) classifier might have better performance compared to NB, nevertheless, it is a hard and time-consuming work to learn the weight values for all features. Cuckoo search is a newly proposed meta-heuristic optimization algorithm which has been successfully applied for many parameter optimization problems. In the paper, a remote image classification approach is proposed, the attribute weight of which is learnt through cuckoo search algorithm (CSWNB in brief). In order to testify the performance of the proposed method, it is compared to some other evolutionary algorithms, such as attributed weighted Naive Bayes based on Genetic Algorithm (GAWNB), attributed weighted Naive Bayes based on Particle Swarm Optimization (PSOWNB) and attributed weighted Naive Bayes based on Water Wave Optimization (WWOWNB) etc. Experimental results demonstrate that the proposed approach has higher classification accuracy and more stable performance.**

Keywords—Image classification; Cuckoo search Algorithm; Weighted Naive Bayes classifier; Optimization

I. INTRODUCTION

Remote sensing image classification is the primary work in the field of remote sensing image information processing, which directly affects the development of remote sensing technology. In remote sensing images, objects belonging to the identical class have the same or similar spectral and spatial information characteristics under the equal conditions. Therefore, the feature vectors of these objects will cluster in the unified feature space regions, while the different objects will cluster by contraries due to the characteristics of spectral or spatial information[1]. Around the topic, many approaches have been put forward, most of the remote sensing image classification methods are based on the statistical analysis of random variables, such as maximum likelihood[2], *k*-nearest neighbor[3], support vector machine(SVM)[4], decision tree[5] and Naive Bayes[6]. In addition, some classifier ensemble methods are also used to improve classification accuracy, such as bagging, adaboost, random forest and rotation forest[7].

Owing to high computational efficiency and accuracy, The Naive Bayes classifier(NB) has been widely used[8-9]. NB is based on the conditional independence hypothesis, that is, the effect of each attribute on given class is independent of other attributes. When the conditional independence assumption is established, Naive Bayes classification algorithm has the smallest possibility of error[10]. Based on the conditional independence assumption, Naive Bayes classification algorithm considers each attribute has the same effect on the class attribute. In fact, some of the attributes have great influence on classification, while some have less. If considering the unrelated, redundant, and noise-abusive properties attributes as equal status as others, the accuracy of classification will decrease. In order to improve the accuracy of NB and expand its applicability, various attribute weighted algorithms and NB are combined. As a result, Naive Bayes is extended to weighted NB that can assign different weights to various feature categories according to its influence, which could be considered as a parameter optimization problem.

In order to find the appropriate weight value of WNB to improve its classification accuracy, the scholars have proposed a number of methods to assess the importance of attributes including Gain Ratio[11], Relief attribute ranking algorithm and correlation-based feature selection algorithm. Chen et al.[12] proposed to utilize association rules to improve the Naive Bayes classification(NBC) algorithm, through the association rule of the confidence. Zhang[13] presented the probability of success of WNB as a weighting factor. To improve the value of the area under the receiver operating characteristic curve, Hall[14] proposed a attribute weighting method based on the degree of dependence on other attributes. However, to learn the weight values of all the feature is an optimization problem with multivariate parameters, which has many the local optimal solutions. Therefore, the optimal solution of the problem could not be obtained by simple gradient descent method.

By observing the great laws of nature, several modern meta-heuristic algorithms are often used to solve various complex optimization problems. Some of these optimization issues include reliability issues, process scheduling, image segmentation, feature selection and parameter optimization. For instance, Jie Lin et al.[15] proposed weighted Naive

Corresponding author weizhiye121@163.com

978-1-5386-3016-7/17/$31.00 ©2017 IEEE

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 29,2026 at 10:46:04 UTC from IEEE Xplore. Restrictions apply.

Bayes classification algorithm based on particle swarm optimization. Wu[16] et al. demonstrated that weighting scheme for WNB based on artificial immune systems(AISWNB) can successfully find out the optimal weight combinations for different learning tasks, and its performance consistently outperforms Naive Bayes algorithms. However, these algorithms are easy to fall into the local optimum.

Inspired by the intelligent hatching behavior of cuckoos in nature, Yang et al. put forward a meta-heuristic optimization algorithm, which is called cuckoo search(CS)[17]. CS has been proved to be reliable efficient while solving real-world problems[18-20]. Civicioglu et al.[21] analyzed CS, particle swarm optimization, differential evolution, and artificial bee colony algorithms on global optimization problems, results shows that CS achieves very robust and precise solutions . The optimization problem of attribute weight can be abstracted as a function optimization problem, which can be solved by Meta-heuristic optimization algorithm. Hence, CSWNB is considered here, and has achieved good performance when applied to the real image classification.

The reminder of the paper is structured as below. A brief overview of attribute weighed Naive Bayes classification algorithm is given in Section 2. Section 3 describes the principle and pseudo-code of the proposed method. Section 4 displays the experimental results and discussion on five public data sets and real texture image data sets. Finally, the paper is concluded in section 5.

II. Attribute Weighted Naive Bayes Classifier Algorithm

Given a training sample $D=\{x_1,x_2,\cdots,x_N\}$ with M instances, each instance contains n attribute values and a class label. $x=\{x_{i,1},\cdots,x_{i,j},\cdots,x_{i,n},y_i\}$ is utilized to stand for the i-th instance of data sample D, with $x_{i,j}$ representing the j-th attribute value and y_i representing the class label of the case. The class space $y=\{c_1,\cdots,c_k,\cdots,c_L\}$ represents the labels to which each instance belongs and c_k represents the k-th label of classifying space. For convenience, (x_i,y_i) is used to express an case and its category label.

Every attribute could be a discrete or continuous random variable. For every attribute A_j, $a_j^{\tau},\tau=1,\cdots,|A_j|$ is utilized to represent the $\tau-th$ attribute value of A_j, while $|A_j|$ represents the sum of different values of A_j. The value of case x_i contents $x_{i,j}\in A_j$. For each instance (x_i,y_i) in training sample D, its category label satisfied $y_i\in Y$ and its category label y_t demands to be divined by Naive Bayes model. In Naive Bayes model, with class label training instances are given to establish Naive Bayes model ahead of time, which might be formulated as Eq.(1):

$$c(x_t)=\arg\max_{c_k\in y}P(c_k)P(x_{t,1},x_{t,2},\cdots,x_{t,n}\mid c_k) \tag{1}$$

In Eq.(1), $P(c_k)$ represents the probability of class c_k among all of the training sample. $P(x_{t,1},x_{t,2},\cdots,x_{t,n}\mid c_k)$ describes the joint distribution of x_t which conditioned by given category c_k. Under the assumption that all attributes are independent, by observing the variables, the probability of observing the connection is only the product of the verifiability of the individual attributes. As a simple but useful model, Naive Bayes is defined as Eq.(2).

$$c(x_t)=\arg\max_{c_k\in y}P(c_k)\prod_{j=1}^{n}P(x_{t,j}\mid c_k) \tag{2}$$

where $P(x_{t,j}\mid c_k)$ represents the joint distribution of $x_{t,j}$, which conditioned by given category c_k.

Given a test case x_t, which contains the values of attribute simply, the class label y_t for each x_t requires to be divined by WNB model which is formulated as Eq.(3)

$$c(x_t)=\arg\max_{c_k\in y}P(c_k)\prod_{j=1}^{n}P(x_{t,j}\mid c_k)^{w_j} \tag{3}$$

where w_j signifies the weight of the attribute j, which is in the scope of $[0,2]$. $P(c_k)$ represents the probability of class c_k among all of the training sample. $P(x_{t,j}\mid c_k)$ has the same meaning as Eq.(2).

III. The Proposed Method

In traditional WNB, each attribute has the same effect on the result. In fact, the importance of each attribute is not the same. In order to optimize WNB, CS is employed to learn the suitable weight value of the attribute. The weight vale is between 0 and 2. CS is applied to search for the optimal solution, that is, the most appropriate weight.

A. The Encoded Scheme

The weight value of the feature is a real number and CS utilizes real-number encoding itself. As a result, it is a very convenient for employing CS to train the weight value for WNB with real number encoding. The candidate weight value is coded with a vector and the encoding length is equal to the number of features. The weight value is set in the range of $[0,2]$.

B. The definition of fitness function

In CS algorithm, fitness is used to distinguish the quality of individual groups, the structure of which directly affects the quality of the solution. When the individual has greater fitness value, its performance is also better. The solving process of CS algorithm is to choose the individual with the fitness value to ensure that the better nest is reserved for the next generation. In this paper, the weighted Naive Bayes classifier is utilized to calculate the classification accuracy as the fitness value. The fitness function of parameter optimization is defined as:

$$f(i)=\frac{True}{True+False} \tag{4}$$

978-1-5386-3016-7/17/$31.00 ©2017 IEEE

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 29,2026 at 10:46:04 UTC from IEEE Xplore. Restrictions apply.

Where *True* indicates the correct number of categories by CSWNB classifier, and *False* is the number of errors. The result of CSWNB classifier is calculated by Eq.(3).

C. The pseudo-code of attribute weighed Naive Bayes classification algorithm based on cuckoo search algorithm

The main process of the proposed method is as follows:

Algorithm 1. Weighed Naive Bayes classification algorithm based on CS algorithm.

Input: Training samples.

Output: Weight values of features.

1. Randomly initialize a population P of n nests (solutions) $x_i (i = 1, 2, ..., n)$;
2. **While** stop criterion is not satisfied **do**
3. Classify the samples and use Naive Bayes to calculate the classification accuracy as the fitness value $f(x)$ for each solution by Eq. (4). Determine the best solution x^*.
4. **for** each $x \in P$ **do**
5. Get a cuckoo randomly by Levy flights;
6. Evaluate its fitness $f(x')$ by Eq. (4);
7. **if** $f(x') > f(x)$ **then**
8. Replace x by the new solution x';
9. **if** $rand_i < p_a$ **then**
10. Nest x_i is abandoned and a new one is built;
11. Keep the best solutions;
12. Rank the solutions and find the current best x^*
13. **end while**
14. return x^*.

IV. Experiment and Discussion

A. Experiments for Public Data Sets

In order to verify the effectiveness and accuracy of the proposed CSWNB algorithm, the data set tested are chosen from UCI machine learning repository, including the data set Segment, Wdbc, Ionosphere, Biodeg and PhishingData. The general information of the data sets is shown in Table I.

In order to improve the reliability of the experimental results, the same training set and testing set are used for different algorithms, accounting for 70% and 30% of the total data respectively. The experiment is divided into two parts. Firstly, the classification accuracy of the five experimental data sets is obtained by using WNB. Then, GAWNB, PSOWNB, WWOWNB and CSWNB are utilized to test on the five data sets and calculate the classification accuracy separately.

The parameters of the algorithm are briefly described. For GAWNB, the number of genetics N, crossover probability p_c, mutation probability p_m and the number of iteration are set to 60, 0.8, 0.05 and 200 respectively. For PSOWNB, the number of particles N, acceleration constant c_1, c_2, inertia coefficient ω and the number of iteration are set to 60, 2, 2, 1 and 200 respectively. For WWOWNB, the number of waves N, the maximum wave height $h_{\max}$, wavelength reduction coefficient α, breaking coefficient β, the maximum number of breaking directions $k_{\max}$ and the number of iteration are set to 60, 12, 1.0026, 0.25, 12 and 200 respectively. For WWOWNB, the number of waves N, the maximum wave height $h_{\max}$, wavelength reduction coefficient α, breaking coefficient β, the maximum number of breaking directions $k_{\max}$ and the number of iteration are set to 60, 12, 1.0026, 0.25, 12 and 200 respectively.

TABLE I. General Information of Public Data Sets From UCI

Data Set	Instance	Attribute	Class
Segment	2310	19	7
Wdbc	569	30	2
Ionosphere	351	34	2
Biodeg	1055	41	2
PhishingData	1353	9	3

(Http://Archive.Ics.Uci.Edu/Ml/Datasets.Html)

For CSWNB, the number of cuckoos N, detecting probability p_a, step size a and the number of iteration are set to 60, 0.3, 10 and 200 respectively. Moreover, to make the results more equitable, the experiment was carried out 10 times for each data set, and finally the maximum(Max), minimum(Min), mean(Mean) and standard deviation(Std.) of the ten results were obtained. The results of the experiment are shown in Table Ⅱ.

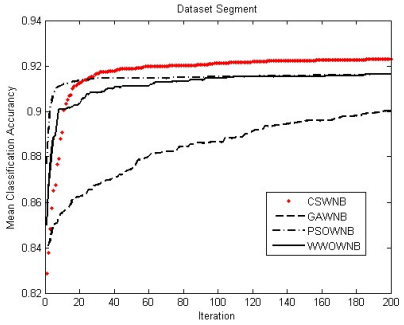
As can be seen from Table Ⅱ, the performance of attribute WNB is significantly better than Naive Bayes algorithm. At the same time, the accuracy of classification using CSWNB is higher than GAWNB, PSOWNB and WWOWNB. For the data set Segment, compared to the Naive Bayes, of which the classification accuracy is 79.3%, the results of attribute weighted algorithm all exceeded 90.0%, where the mean classification accuracy and maximum of CSWNB is the highest, reaching 92.3% and 92.6% respectively. Similarly, for the data set Biodeg, the mean classification accuracy of CSWNB is 85.3%, higher than GAWNB, PSOWNB and WWOWNB, which are respectively 77.9%, 82.0% and 80.0%. It can be clearly seen that CSWNB has better performance than Naive Bayes with 15% higher correct rate. For other data sets, such as Wdbc, Ionosphere and PhishingData, the mean classification accuracy of CSWNB reaches 98%, 97.9% and 82.8%, which is better than other attribute weighted Naive Bayes classifier. Likewise, convergence learning curves can intuitively demonstrate experimental results for five public data sets, shown in Fig.1. In order to further study the convergence of CSWNB algorithm and intuitively demonstrate the experimental results, convergence learning curves have been drawn to show the change rule between

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 29,2026 at 10:46:04 UTC from IEEE Xplore. Restrictions apply.

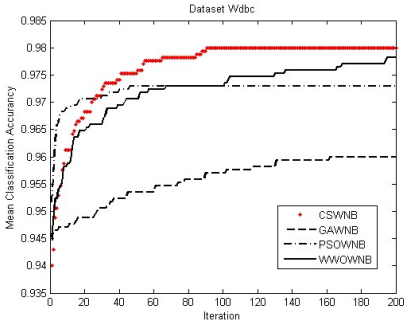
classification accuracy and the number of iterations for five public data sets, which is shown in Fig.1. Each point on the curve is the average of ten experimental results. It can be seen that the CSWNB algorithm has better performance in jumping out of the local search and obtaining higher classification accuracy than other four algorithms.

TABLE II. TABLE COMPARISON OF CLASSIFICATION ACCURACY FOR FIVE PUBLIC DATA SETS

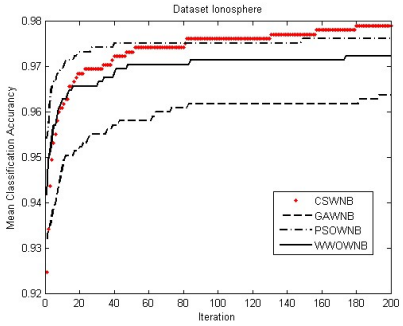
Data set	Accuracy	Naive Bayes	GAWNB	PSOWNB	WWOWNB	CSWNB
Segment	Mean	79.30%	90.00%	91.60%	91.60%	92.3%*
	Max		91.50%	91.90%	91.80%	92.60%
	Min		87.70%	91.30%	91.50%	92.10%
	Std.		0.015	0.002	0.001	0.002
Wdbc	Mean	91.80%	96.00%	97.30%	97.80%	98.0%*
	Max		97.10%	98.20%	98.20%	98.20%
	Min		95.30%	95.90%	97.60%	97.60%
	Std.		0.004	0.007	0.003	0.003
Ionosphere	Mean	87.60%	96.40%	97.60%	97.20%	97.9%*
	Max		97.10%	98.10%	98.10%	98.10%
	Min		95.20%	97.10%	97.10%	97.10%
	Std.		0.006	0.005	0.003	0.004
Biodeg	Mean	69.60%	77.90%	82.00%	80.00%	85.3%*
	Max		79.40%	85.80%	81.60%	86.40%
	Min		76.30%	78.50%	78.80%	83.90%
	Std.		0.90%	2.10%	0.80%	0.60%
PhishingData	Mean	78.30%	82.00%	81.90%	82.40%	82.8%*
	Max		82.50%	82.50%	82.70%	83.00%
	Min		81.20%	81.20%	82.20%	82.50%
	Std.		0.003	0.004	0.001	0.002



(A)SEGMENT

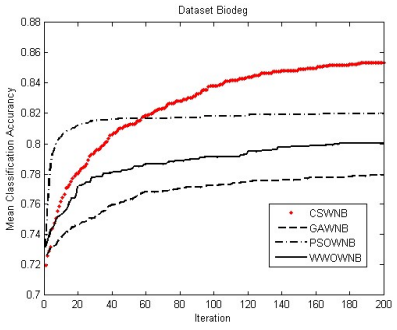


(B)WDBC

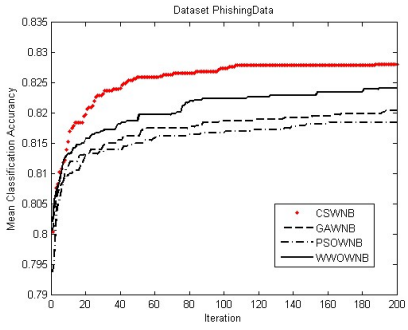


(C)IONOSPHERE

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 29,2026 at 10:46:04 UTC from IEEE Xplore. Restrictions apply.



(D)BIODEG



(E)PHISHINGDATA

FIGURE 1 CONVERGENCE LEARNING CURVES OF CSWNB, GAWNB, PSOWNB AND WWOWNB FOR FIVE PUBLIC DATA SET. (A)SEGMENT (B)WDBC. (C)IONOSPHERE. (D)BIODEG. (E)PHISHINGDATA.

B. Experiments for Real Image Data Sets

To further illustrate the performance of the proposed method on real image data sets, two real remote sensing image data sets are selected as training data in this part. Image data set_1 has 22 attributes and Image data set_2 has 78 attributes, which are divided into four classes, including resident, vegetation area, water and paddy field. Some training samples are shown in Fig.2.

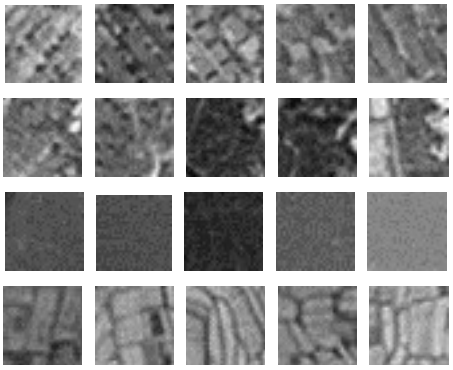


FIGURE 2 SOME TRAINING SAMPLE

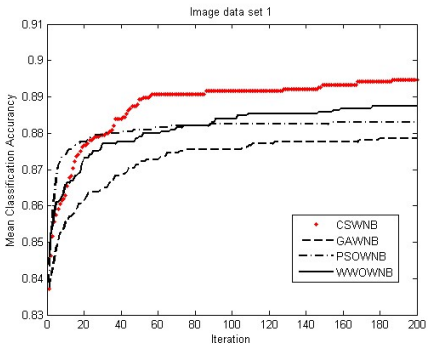
The corresponding results are displayed in Table III.

As can be observed from Table III, for the first image data set, the mean accuracy of CSWNB is higher than Naive Bayes, GAWNB, PSOWNB and WWOWNB, reaching 89.5%, and the maximum of classification accuracy of CSWNB and PSOWNB has the same value 89.8%, which is better than GAWNB and WWOWNB. For the second image data set, the maximum, minimum and mean classification accuracy of CSWNB are respectively 100%, 99.7% and 99.9%. It means CSWNB can almost classify correctly all testing images in 10 experiments. The mean classification accuracy of PSOWNB and WWOWNB are respectively 99.3% and 99.4%, which are both lower than CSWNB. It is might deduce that WNB is clearly better than Naive Bayes algorithm. In the same way, convergence learning curves are utilized to demonstrate experimental results for two image data sets, shown in Fig.3. It is watched that from Fig.3, CSWNB algorithm has faster convergence rate and better perform in jumping out of the local optima than other three algorithms for two image data sets.

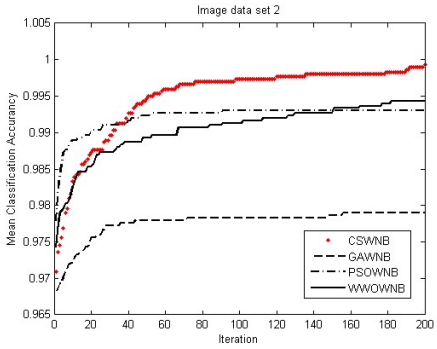
TABLE III. COMPARISON OF CLASSIFICATION ACCURACY FOR IMAGE DATA SETS

Data set	Accuracy	Naive Bayes	GAWNB	PSOWNB	WWOWNB	CSWNB
Image data set_1	Mean	80.00%	87.9%	88.3%	88.7%	89.5%*
	Max		89.3%	89.8%	89.3%	89.8%
	Min		86.3%	86.8%	88.3%	88.8%
	Std.		0.008	0.009	0.003	0.003
Image data set_2	Mean	97.8%	97.9%	99.3%	99.4%	99.9%*
	Max		99.0%	100.0%	99.7%	100.0%
	Min		97.0%	98.0%	99.0%	99.7%
	Std.		0.005	0.006	0.003	0.001

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 29,2026 at 10:46:04 UTC from IEEE Xplore. Restrictions apply.



(A)IMAGE DATA SET_1



(B)IMAGE DATA SET_2

FIGURE 3 CONVERGENCE LEARNING CURVES OF CSWNB, GAWNB, PSOWNB AND WWOWNB FOR IMAGE DATA SET.(A)IMAGE DATA SET_1(B)IMAGE DATA SET_2

V. CONCLUSIONS

In this paper, cuckoo search algorithm is applied for weighted Naive Bayes classification. In CSWNB, the weighting coefficient is more reasonable according to the effects of the different values on the classification results. From the classification results of the five data sets in the UCI database and the real image data sets, it is verified that the proposed algorithm has good performance, which can improve the classification accuracy, and the result is more stable, especially in the case of classifying the data sets with high redundancy attributes.

ACKNOWLEDGMENT

This work is funded by the National Natural Science Foundation of China under Grant No.41301371, 51372076, 61502155, 6177051856, funded by State Key Laboratory of Geo-Information Engineering, No. SKLGIE2014-M-3-3 and supported by project of Hubei Provincial Department of Education (Q20131407).

REFERENCES

[1] Cui S, Schwarz G, Datcu M. Remote Sensing Image Classification: No Features, No Clustering[J]. IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing, 2016, 8(11):5158-5170.

[2] Sun J, Yang J, Zhang C, et al. Automatic remotely sensed image classification in a grid environment based on the maximum likelihood method[J]. Mathematical & Computer Modelling, 2013, 58(3-4):573-581.

[3] Li Y, Cheng B. An improved k-nearest neighbor algorithm and its application to high resolution remote sensing image classification[J]. 2009:1-4.

[4] Zhou J, Qin J, Gao K, et al. SVM-based soft classification of urban tree species using very high-spatial resolution remote-sensing imagery[J]. International Journal of Remote Sensing, 2016, 37(11):2541-2559.

[5] Yang G, Fang S. Improving remote sensing image classification by exploiting adaptive features and hierarchical hybrid decision trees[J]. Remote Sensing Letters, 2017, 8(2):156-164.

[6] Sitthi A, Nagai M, Dailey M, et al. Exploring Land Use and Land Cover of Geotagged Social-Sensing Images Using Naive Bayes Classifier[J]. Sustainability, 2016, 8(9):921.

[7] Zhang H, Ren J X, Kang Y L, et al. Development of novel in silico model for developmental toxicity assessment by using naïve Bayes classifier method.[J]. Reproductive Toxicology, 2017:8-15.

[8] Xia J, Du P, He X, et al. Hyperspectral Remote Sensing Image Classification Based on Rotation Forest[J]. IEEE Geoscience & Remote Sensing Letters, 2013, 11(1):239-243.

[9] Hameg S, Lazri M, Ameur S. Using Naive Bayes classifier for classification of convective rainfall intensities based on spectral characteristics retrieved from SEVIRI[J]. Journal of Earth System Science, 2016, 125(5):1-11.

[10] Krakovna V, Du J, Liu J S. Selective Bayesian Forest Classifier: Simultaneous Feature Selection and Classification[J]. Statistics, 2015.

[11] H. Zhang and S.L. Sheng, "Learning weighted naive Bayes with accurate ranking" in Proceedings of the 4th IEEE International Conference on Data Mining, ICDM04, IEEE Computer Society, USA, 2004, pp.567-570.

[12] Chen C D, Liang Z X, Zheng S J. Improved Naive Bayesian classification algorithm based on association rules [J]. Journal of Computer Applications, 2010, 19 (11): 106-109.

[13] Zhang H, Sheng S. Learning weighted naive Bayes with accurate ranking. 2004 Fourth IEEE International Conference on. Data Mining(ICDM' 04). IEEE. 2004. 567-570.

[14] Hall M. A Decision Tree-Based Attribute Weighting Filter for Naive Bayes[J]. Knowledge-Based Systems, 2007, 20(2):120-126.

[15] Lin J, Yu J. Weighted Naive Bayes classification algorithm based on particle swarm optimization[C]// IEEE, International Conference on Communication Software and Networks. IEEE, 2011:444-447.

[16] Wu J, Cai Z, Zeng S, et al. Artificial immune system for attribute weighted Naive Bayes classification[C]// International Joint Conference on Neural Networks. IEEE, 2013:1-8.

[17] Yang X S, Deb S. Cuckoo Search via Levy Flights[J]. Mathematics, 2010:210 - 214.

[18] Nadjemi O, Nacer T, Hamidat A, et al. Optimal hybrid PV/wind energy system sizing: Application of cuckoo search algorithm for Algerian dairy farms[J]. Renewable & Sustainable Energy Reviews, 2017, 70:1352-1365.

[19] Kout A, Labed S, Chikhi S, et al. AODVCS, a new bio-inspired routing protocol based on cuckoo search algorithm for mobile ad hoc networks[J]. Wireless Networks, 2017:1-11.

[20] Sun W, Sun J. Daily PM2.5 concentration prediction based on principal component analysis and LSSVM optimized by cuckoo search algorithm[J]. Journal of Environmental Management, 2017, 188:144.

[21] Civicioglu P, Besdok E. A conceptual comparison of the Cuckoo-search, particle swarm optimization, differential evolution and artificial bee colony algorithms[J]. Artificial Intelligence Review, 2013, 39(4):315-346

Authorized licensed use limited to: BMS College of Engineering. Downloaded on October 29,2026 at 10:46:04 UTC from IEEE Xplore. Restrictions apply.